

# Joshua Greenhalgh

SENIOR DATA & SOFTWARE ENGINEER

United Kingdom

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## Summary

Senior data and software engineer with ten years of experience building production systems across data platforms, cloud infrastructure, machine learning and backend engineering. A strong generalist who moves comfortably between system design, Python development, data modelling, orchestration and infrastructure, and who is used to owning technically ambiguous projects from initial exploration through architecture, implementation and production operation. Builds reliable, maintainable systems for complex and high-volume data with a focus on observability, testing and reproducibility. Actively builds with PyTorch and LLMs outside work, from interpretability tooling to agentic workflows. Brings strong mathematical and computational foundations, clear communication and pragmatic engineering judgement, and is most effective in small teams where autonomy, broad technical ownership and close collaboration matter.

## Experience

### Senior Software Engineer

UK

IMU BIOSCIENCES

Feb 2024 – Present

- Designed and built the **data platform and catalogue** for 100k+ single-cell (flow cytometry) samples, unifying scattered Parquet and vendor file sources into a structured, lineage-aware index. Cut query times and made ML training datasets fully reproducible.
- Owned **storage architecture** for high-volume single-cell data end to end: benchmarked Zarr and Parquet access patterns across S3 and Snowflake and delivered a hybrid design serving both whole-sample retrieval and cross-sample aggregation for model training.
- Built a **QC and monitoring framework** in Dagster that automatically validates reference-sample consistency across experiments, catching data-quality regressions before they reach analysts or models.
- Defined **domain ontologies and metadata standards** with immunologists and embedded them in pipeline outputs, standardising labelling across diverse assays and improving the interpretability of downstream ML models.
- Developed a **synthetic data generation framework** that produces realistic single-cell datasets, used to stress-test pipelines and validate ML model generalisation.

### Sabbatical

Spain

CAREER BREAK

Oct 2023 – Jan 2024

- Planned time off between roles; walked the Camino de Santiago.

### Senior Data Engineer

UK

MOBKOI

Jan 2022 – Oct 2023

- Rebuilt the company's data pipelines from **ad-hoc cloud functions to Prefect orchestration** on GCP with CI/CD, increasing deployment speed and safety while reducing operational overhead.
- Led migration from aggregated partner data to **impression-level ingestion** from AWS ad-serving infrastructure and CloudWatch logs, enabling granular analytics and increasing client confidence in reported numbers.
- Designed and implemented a **new data warehouse** with change data capture from internal operational systems, improving data freshness and reliability for the analytics team.
- Overhauled **revenue estimation and FX handling**, reducing monthly variance against P&L from ~20% to under 1%.
- Introduced **infrastructure as code and version control** across the organisation, migrating all source to GitHub and making data workflows and deployments visible and auditable.

### Founder

UK

INDEPENDENT VENTURE

Apr 2021 – Dec 2021

- Left full-time employment to start a business. Owned everything from product and technical build to customer discovery; wound the venture down when it did not find market fit and returned to engineering.

### Senior Data Engineer

UK

TRANSITIONZERO

Jan 2021 – Mar 2021

- Implemented **Terraform-managed infrastructure on GCP** with revision-based deployments, and replaced Cloud Composer with **Prefect pipelines** and a CI-based framework that let analysts deploy their own pipelines independently.
- Built satellite imagery processing pipelines on **Google Earth Engine** feeding vision-based ML models for climate analytics.

Senior Data Engineer

HEYJOBS

Berlin, Germany

Aug 2019 – Nov 2020

- Built a **data quality framework for the Redshift warehouse** using property-based testing of SQL transformations, integrated with Airflow, DataDog and Slack as part of CI. Significantly reduced post-deployment data issues.
- **Tech lead** for redesigning marketing cost processing, migrating undocumented legacy SQL to fully unit-tested Python and reducing marketing overspend by ~5%.
- Established **Airflow best practices** (incremental, idempotent, deterministic pipelines) and moved intermediary data to AWS Glue and Spectrum, cutting Redshift storage costs.

Data Engineer / Analyst

FLOWMINDER FOUNDATION

Southampton, UK & Berlin (remote)

Jun 2017 – Jun 2019

- **Led a GSMA-funded project end to end**, including budget and contractors, standardising ETL on **Airflow** across multiple countries (50M+ mobile phone records ingested daily) and building a post-crisis mobility analysis tool on Kafka and TimescaleDB for humanitarian aid targeting.
- Led an **Inter-American Development Bank** project estimating commuting flows in Haiti's two largest cities from three years of mobile phone data (~10M subscribers), delivering interactive PostGIS and React dashboards to the transport ministry.
- Designed and delivered a two-week **Python and R training programme** for Nepal's National Statistics Office on JupyterHub (GKE).

Doctoral Researcher, Computational Modelling

UNIVERSITY OF SOUTHAMPTON

Southampton, UK

Sep 2015 – Jun 2017

- Researched **deep learning, optimisation and Bayesian methods** for inverse problems in x-ray tomography (TensorFlow, PyMC3, OpenMP/MPI on HPC). Taught undergraduate Python and mathematics and postgraduate research software engineering. Left to accept a permanent offer from Flowminder after an internship there.

Projects

SELECTED PERSONAL ML & LLM PROJECTS, OPEN SOURCE AT GITHUB.COM/JOSH-GREE

2025 – 2026

- **probe-lab**: toolkit for LLM activation probing. Extracts per-layer residual-stream activations with PyTorch and Hugging Face Transformers and fits linear probes against prompt metadata for interpretability experiments.
- **weight-corruption**: experimental framework measuring how weight perturbations (noise, zeroing, shuffling, quantisation) degrade language modelling and behaviour across the Pythia model family, with GPU evaluation on Modal.
- **birdclef-2026** and **wm**: bird-audio classification in PyTorch (timm, ONNX embeddings) trained on Modal GPUs with Weights & Biases tracking; wrote **wm**, a small framework that snapshots every experiment run as a PR for reproducibility.
- **news-knowledge-graph**: LLM extraction pipeline turning news articles into a knowledge graph using DSPy programs and GLiNER2 for entity extraction, temporal tagging and Wikidata entity linking, orchestrated with Prefect.
- **mobile-agents**: GitHub Actions workflow built on the Claude Agent SDK that plans and implements issues autonomously and opens pull requests.

Skills

|                           |                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Languages                 | Python, SQL, R, JavaScript                                                                                       |
| Orchestration & Pipelines | Dagster, Prefect, Airflow, dbt, Kafka, change data capture, ETL/ELT, CI/CD                                       |
| Storage & Warehousing     | Snowflake, Redshift, Postgres, PostGIS, pgvector, TimescaleDB, Parquet, Zarr, S3, data modelling                 |
| Cloud & Infrastructure    | GCP, AWS, Terraform, Docker, Kubernetes, GitHub Actions, infrastructure as code                                  |
| ML & AI                   | PyTorch, Hugging Face Transformers, TensorFlow/Keras, DSPy, Claude Agent SDK, RAG, Weights & Biases, Modal, PyMC |
| Quality & Testing         | pytest, property-based testing, data quality monitoring, DataDog, reproducible pipelines                         |
| Backend & Web             | REST APIs, Flask, React, JupyterHub                                                                              |

Education

MSc Computational Modelling

UNIVERSITY OF SOUTHAMPTON

Southampton, UK

2014 – 2015

PGCE Mathematics and Qualified Teacher Status

MANCHESTER METROPOLITAN UNIVERSITY

Manchester, UK

2012 – 2014

BSc Mathematics, First Class Honours

MANCHESTER METROPOLITAN UNIVERSITY

Manchester, UK

2009 – 2012